

YUHE WU(吴宇赫)



Phone: (+86) 152-7916-1860 · Email: yuhewu@hkust-gz.edu.cn · [Homepage](#)

EDUCATION

Hong Kong University of Science and Technology (Guangzhou) , Ph.D. in Fintech	2026.09–Present
Dongbei University of Finance and Economics , Bachelor in Economic Statistics	2022.09–2026.06

A BRIEF INTRODUCTION

I am a **Ph.D. student** in Fintech at the **Hong Kong University of Science and Technology (Guangzhou)**, supervised by Prof. Guang Zhang. I received my B.S. in Economic Statistics from Dongbei University of Finance and Economics (DUFE), where I was supervised by Prof. Zhuang Liu and Prof. Xu Qiang. My research focuses on **LLM capabilities**, spanning **LLM evaluation**, **LLM post-training**, and **generative agents**. I have published papers at venues including **ACL 2026**, **EMNLP 2026**, **KDD 2025**, and **Annals of Operations Research**, with additional manuscripts under review at **NeurIPS 2026**, **ICLR 2027**, **ICASSP 2027**, **AAAI 2027**, and **Management Science**. I also serve as a reviewer for leading venues such as **ACL**, **KDD**, **NeurIPS**, **AAAI**, and **IJOC**.

RESEARCH PUBLICATIONS¹

LLM Evaluation

This line focuses on evaluating and interpreting large language models beyond traditional benchmarks, particularly the processes that shape where factuality and reasoning fail. It studies three connected dimensions: where in generation a failure sits, why it fails, and how that understanding can meaningfully advance the next training step. PRISM probes hallucinations across memory, instruction, and reasoning. BizCompass measures whether disciplinary knowledge becomes performance in a business role. Caught in the Story observes failures in multi-turn conversation and attributes them to a specific stage of post-training. MIND benchmarks long-form value measurement across public events and value dimensions, where a post mixes background with a few stance-bearing sentences; TEF fuses model judgments on MIND by the evidence each sentence carries, rather than by a vote over sentences.

- **Yuhe Wu**, Guang Zhang, Zhuang Liu, et al. *PRISM: Probing Reasoning, Instruction, and Source Memory in LLM Hallucinations*. In **ACL 2026 Main Conference** (CCF-A, CORE A*).
- Jianing Hao*, **Yuhe Wu***, Yuanjian Xu*, Guang Zhang, et al. *BizCompass: Benchmarking the Reasoning Capabilities of LLMs in Business Knowledge and Applications*. In **ACL 2026 Findings** (CCF-A, CORE A*; *Equal contribution).
- **Yuhe Wu**, Guang Zhang, Zhuang Liu, et al. *Caught in the Story: Narrative Captivity in Multi-turn LLMs Conversation*. In **EMNLP 2026 Findings** (CCF-B, CORE A).
- **Yuhe Wu***, Rui Qian*, Guangyu Wang*, Guang Zhang, et al. *Count Evidence, Not Sentences: Tempered Evidence Fusion of LLM Judgments for Long-Text Value Measurement*. **Under review at ICASSP 2027** (CCF-B; *Equal contribution).

LLM Post-Training

Post-training here centers on adaptation of language models—making them precisely steerable, and more capable without making them less reliable. Supervision in this setting is always imperfect; preference and reward signals shape behavior, and self-improving post-training is how that steering is kept. COR² shapes the reward around the structure of an optimization model rather than its outcome. Notation, Not Narration keeps reasoning steerable by retaining only steps a solver can check. BizSage is self-improving post-training for automated research. Continuous integration keeps self-refinement from making an agent less reliable.

- **Yuhe Wu**, Guang Zhang, et al. *BizSage: A Self-Evolving Multi-Agent Framework for Business Research with Efficient Knowledge Retrieval*. In **EMNLP 2026 Findings** (CCF-B, CORE A).
- **Yuhe Wu**, Zhuang Liu, et al. *Enhancing Financial Decision-Making under Cyber Threats: A Dual-Branch Framework Integrating Bayesian Deep Learning and Explainable AI*. In **Annals of Operations Research** (ABS 3, JCR Q1).

¹ ● Published; ● Under review; ● Ongoing

- **Yuhe Wu**, Yuran Chen, Xinyue Su, Zhuang Liu. *PolluVCCT: Multi-Scale Hybrid Learning for Robust Air Pollution Forecasting Across Diverse Climate Zones*. In **KDD-UMC 2025** (CCF-A, CORE A*).
- Xinjian Yi, Rui Qian, Xutao Mao, et al., **Yuhe Wu**[†]. *Continuous Integration for Self-Evolving Agents That Rewrite Their Own Skills*. **Under review at ICLR 2027** (CORE A*; [†]Corresponding author).
- **Yuhe Wu**, Guang Zhang, et al. *Spend Reasoning Where It Counts: Concise LLM Operations Research Modeling*. **Under review at AAAI 2027** (CCF-A, CORE A*).
- **Yuhe Wu**, Guang Zhang, et al. *Notation, Not Narration: Compressing Chain-of-Thought into Solver-Checkable Assertions*. **Target: ACL 2027** (CCF-A, CORE A*).

Generative Agents

The work studies how LLM-based agents perceive, reason, interact, and adapt in multi-agent environments, and how such simulations can be tested against how people perceive, decide, and act, toward systems that are adaptive, reliable, and socially aware. SEDM aligns how agents make decisions in social interaction, rather than the surface answer. Cognitive Boundary probes assumptions about how people decide as structural complexity grows. Cast² simulates collective behavior from the content that starts it. Headwater is a diagnostic method for who failed, by which pathology, and from which round.

- Jiaxin Liu^{*}, **Yuhe Wu**^{*}, Guang Zhang, et al. *Mechanism Alignment for Socially Embedded Decision Making in LLMs*. In **EMNLP 2026 Main Conference** (CCF-B, CORE A; ^{*}Equal contribution).
- **Yuhe Wu**, Guang Zhang, Zhuang Liu, et al. *The Cognitive Boundary of Generative Agents: Economic Coherence under Structural Complexity*. **Under review at Management Science** (UTD24, FT50).
- Guangyu Wang^{*}, **Yuhe Wu**^{*}, Zhaonan Wang. *Beyond When: How Root Content Propagates as Information Cascade Trees*. **Under review at NeurIPS 2026** (CCF-A, CORE A*; ^{*}Equal contribution).
- **Yuhe Wu**, Guang Zhang, et al. *What’s in the Water: Auditing Faults in LLM Social Simulations*. **Target: ACL 2027** (CCF-A, CORE A*).

RESEARCH EXPERIENCE

Fintech Thrust, HKUST(GZ) , Research Assistant, Supervised by Prof. Guang Zhang	Jun 2025 – Aug 2026
National Accounting Center, DUFE , Group Core Member, Supervised by Prof. Xu Qiang	Jan 2025 – Jul 2026
Fintech Lab, DUFE , Group Leader, Supervised by Prof. Zhuang Liu	Mar 2024 – Jul 2026

COMPETITION & GRANTS

• Top 100, ICBC Cup National Fintech Innovation Competition	2025
• National Second Prize, 9th Financial Innovation & Entrepreneurship Competition	2024
• National Third Prize, 10th Student Statistical Modeling Competition	2024
• Core Member, Liaoning Marine Economy Study (5th National Economic Census Project)	2024
• Core Member, Text–Price Correlation in Food Industry (Liaoning Edu. Dept. Innovation Grant)	2024

HONORS & AWARDS

• ACL 2026 Diversity & Inclusion Grant (TOP 1%)	2026
• KDD-25 Undergraduate Scholarship (TOP 1%)	2025
• Dashang Group Academic Research Scholarship (TOP 1%)	2025
• Excellent Innovation and Entrepreneurship Team Scholarship (TOP 1%)	2025
• Advanced Individual in Academic Competitions (TOP 1%)	2025
• First-Class Scholarship (TOP 2%)	2023 & 2024

SKILLS

Machine Learning: LLM post-training (supervised fine-tuning, RLHF, preference optimization), generative models, and agent frameworks; hands-on with the DeepSeek-R1, Qwen, and Llama series.

Statistical & Econometric Analysis: Econometric modeling in SPSS, Stata, SAS, and R; Python and PyTorch for data science and deep learning research.

Research Tools: L^AT_EX, Linux, Conda, Git, and visualization with Origin and Visio.